9 Morphological Image Processing

- The word Morphology commonly denotes a branch of biology that deals with the form and structure of animals and plants
- We use mathematical morphological as a tool for extracting image components that are useful in the representation and description of region shape, such as boundaries extraction, skeletons, convex hull, morphological filtering, thinning, pruning.

Preliminaries

- Reflections

The reflection of a set B, denoted B^{μ}, is defined as

$$B^{\mu} = \{w \mid w = -b, \text{ for } b \in B\}$$

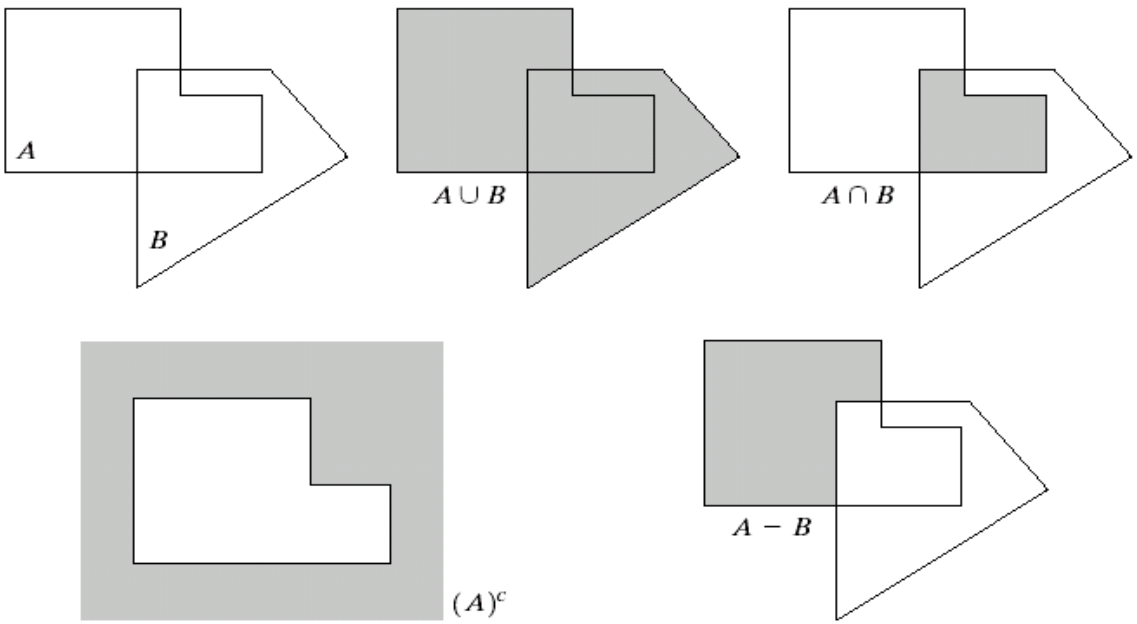
- Translations

The translation of a set B by point $z = (z_1, z_2)$, denoted $(B)_Z$, is defined as

$$(B)_Z = \{c \mid c = b + z, \text{ for } b \in B\}$$

Preliminaries

Operators by examples:



a b c
d e

FIGURE 9.1
(a) Two sets A and B. (b) The union of A and B. (c) The intersection of A and B. (d) The complement of A. (e) The difference between A and B.

Preliminaries

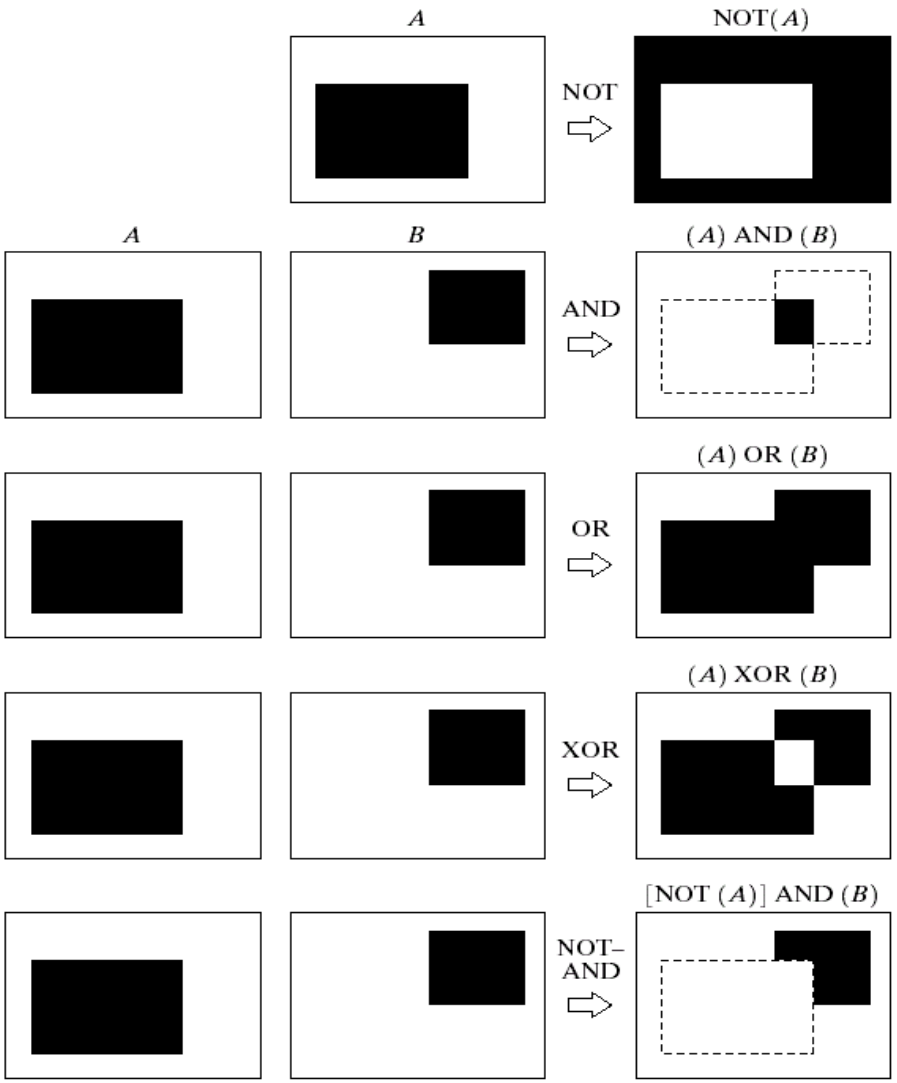
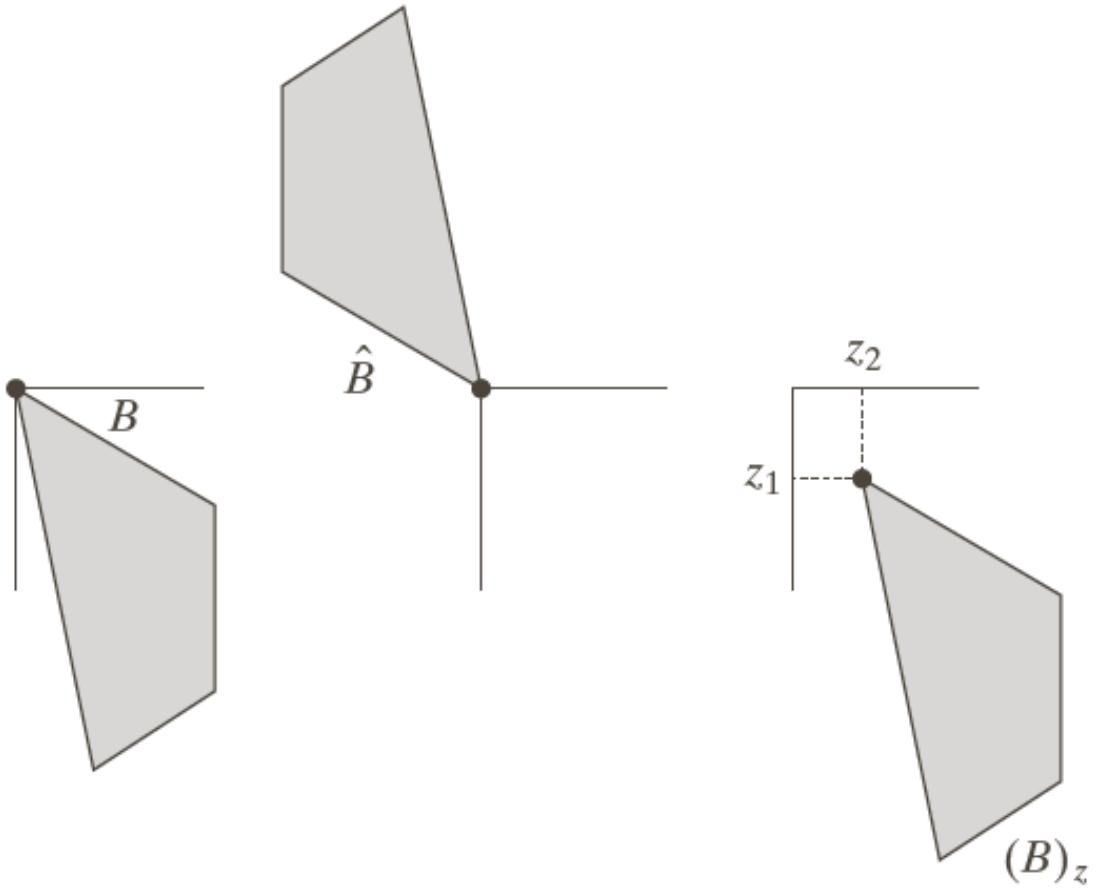


FIGURE 9.3 Some logic operations between binary images. Black represents binary 1s and white binary 0s in this example.

Examples of Reflection and Translation



a b c

FIGURE 9.1
(a) A set, (b) its reflection, and (c) its translation by z.

Structuring Elements

- Small sets or sub-images used to probe an image under study for properties of interest is called SE

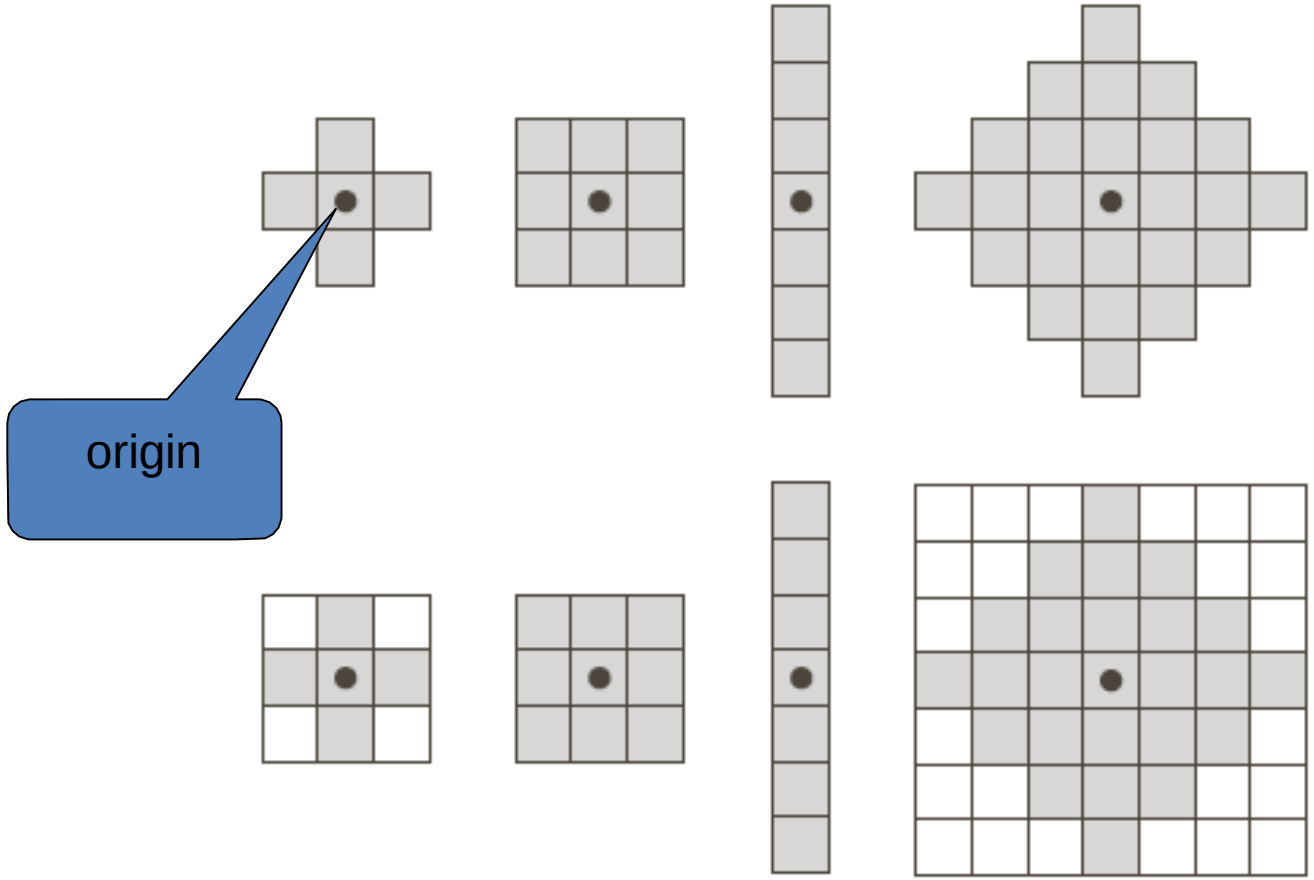
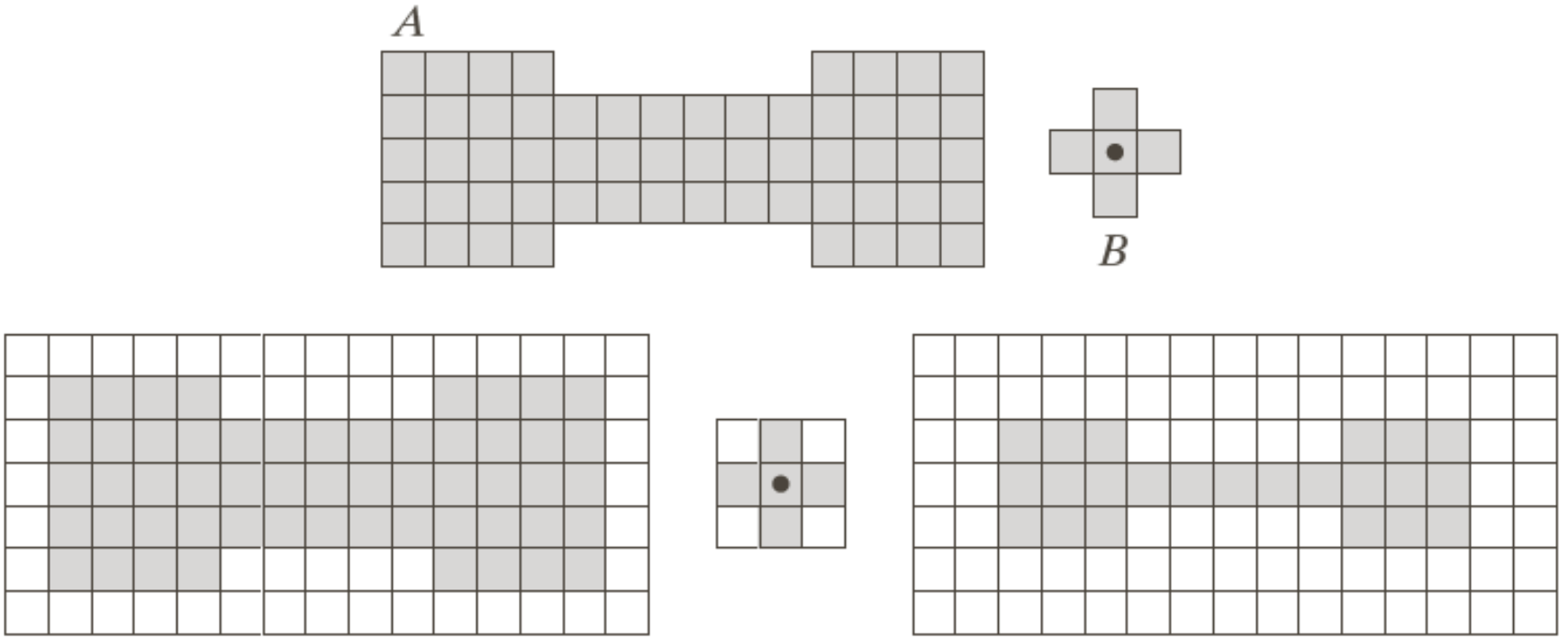


FIGURE 9.2 First row: Examples of structuring elements. Second row: Structuring elements converted to rectangular arrays. The dots denote the centers of the SEs.

Structuring Elements

- The background border is made large enough to accommodate the entire structuring element when its origin is on the border of the original set (Padding)
- SE is of size 3×3 with the origin in the center, so as a one-element border that encompasses the entire set is sufficient



Erosion

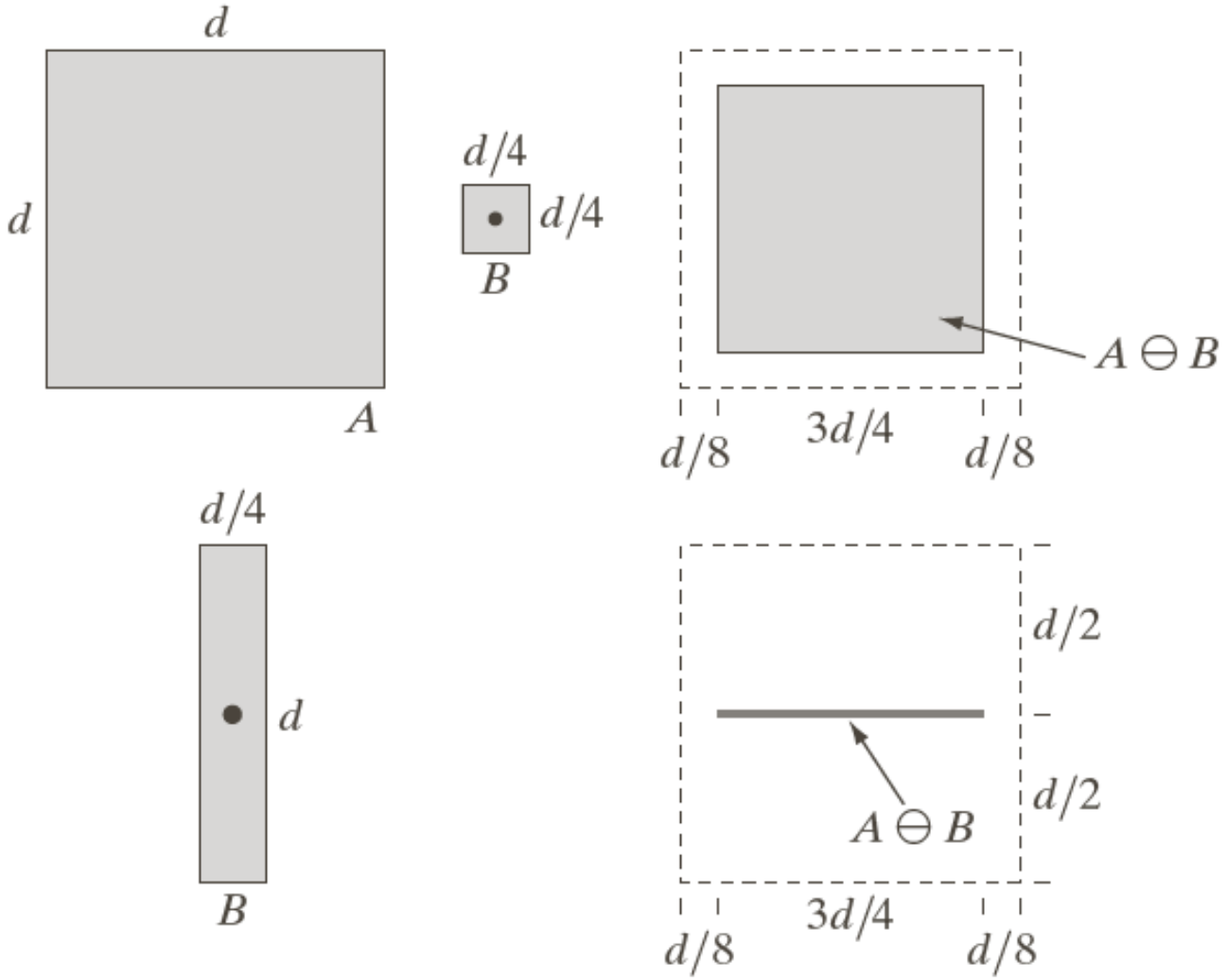
- With A and B as sets in Z^2 , the erosion of A by B is defined as

$$A \ominus B = \{z \mid (B)_Z \subseteq A\}$$

- Erosion of A by B is the set of all points z such that B, translated by z, is contained in A.

$$A \ominus B = \{z \mid (B)_Z \cap A^c = \varnothing\}$$

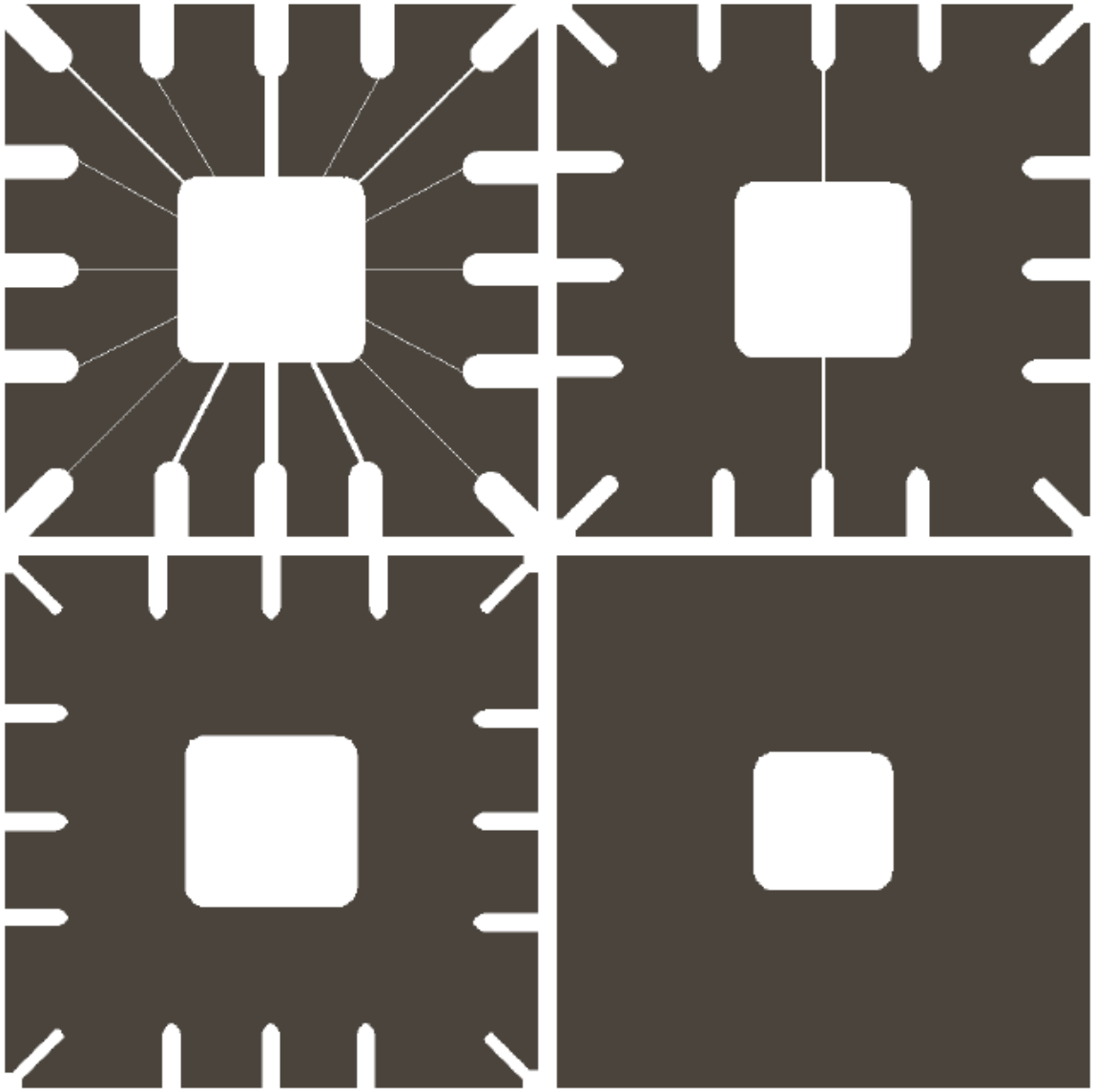
- B has to be contained in A is equivalent to B not sharing any common elements with the background



a b c
d e

FIGURE 9.4 (a) Set A. (b) Square structuring element, B. (c) Erosion of A by B, shown shaded. (d) Elongated structuring element. (e) Erosion of A by B using this element. The dotted border in (c) and (e) is the boundary of set A, shown only for reference.

Erosion



a b
c d

FIGURE 9.5 Using erosion to remove image components. (a) A 486 × 486 binary image of a wire-bond mask. (b)–(d) Image eroded using square structuring elements of sizes 11 × 11, 15 × 15, and 45 × 45, respectively. The elements of the SEs were all 1s.

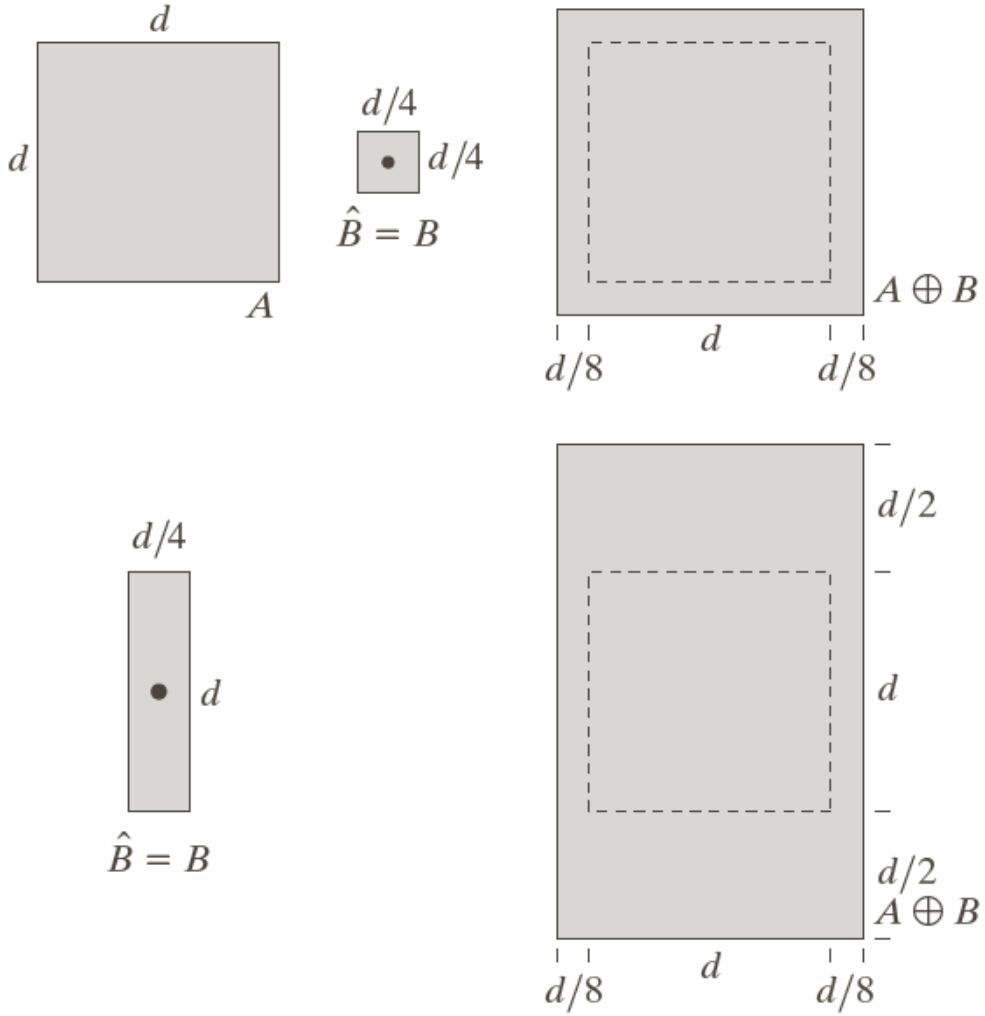
Erosion

- Erosion shrinks or thins objects in a binary image
- Erosion as a morphological filtering operation in which image details smaller than the structuring elements are filtered from the image
- Erosion performed the function of a “line filter”

Dilation

- With A and B as sets in Z^2, the dilation of A by B is defined as $A \oplus B = \left\{ z \mid \left(B^{\mu} \right)_z \cap A \neq \varnothing \right\}$
- Reflecting B about its origin, and shifting this reflection by z
- The dilation of A by B then is the set of all displacements, z, such that B^{μ} and A overlap by at least one element $A \oplus B = \left\{ z \mid \left[\left(B^{\mu} \right)_z \cap A \right] \subseteq A \right\}$

Erosion and Dilation



a b c
d e

FIGURE 9.6
(a) Set A. (b) Square structuring element (the dot denotes the origin). (c) Dilation of A by B, shown shaded. (d) Elongated structuring element. (e) Dilation of A using this element. The dotted border in (c) and (e) is the boundary of set A, shown only for reference

Dilation

- Unlike erosion, dilation “grows” or “thickens” objects in a binary image
- The specific manner and extent of this thickening is controlled by the shape of the structuring element used

Erosion and Dilation





0	1	0
1	1	1
0	1	0

a c
b

FIGURE 9.7
(a) Sample text of poor resolution with broken characters (see magnified view). (b) Structuring element. (c) Dilation of (a) by (b). Broken segments were joined.

Opening and Closing

- Opening of set A by SE B is defined

$$A \circ B = (A - B) \oplus B$$

- Thus, opening A by B is the erosion of A by B, followed by a dilation of the result by B
- Closing of set A by SE B is defined

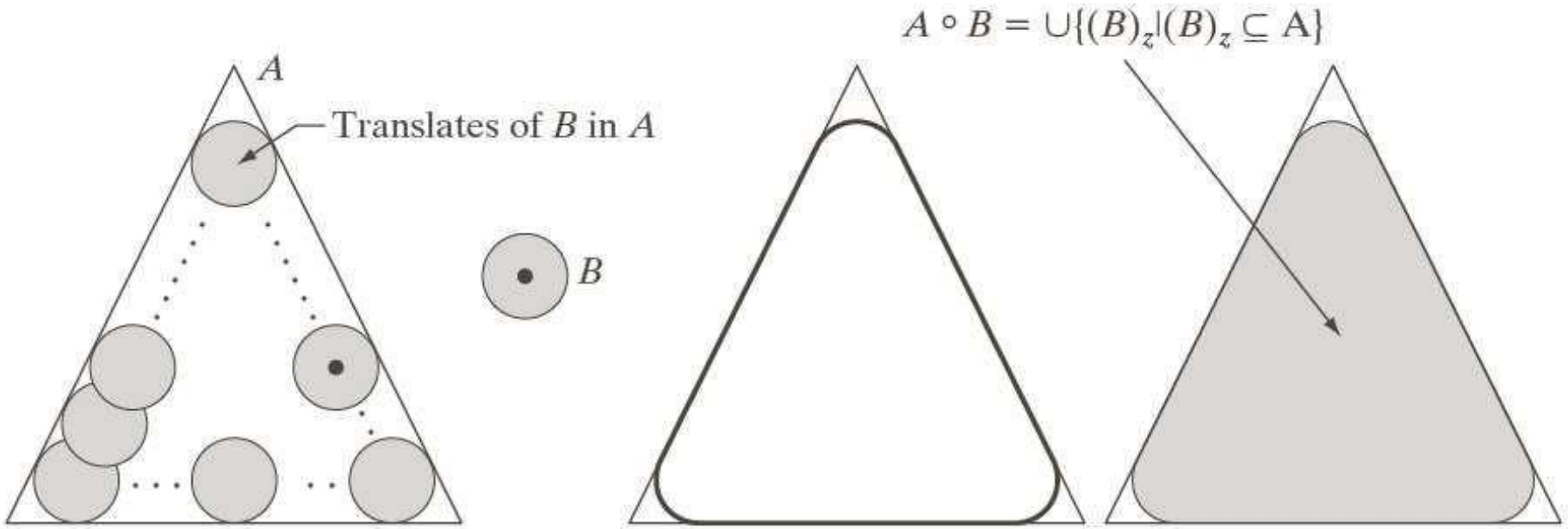
$$A \cdot B = (A \oplus B) \ominus B$$

- Closing of set A by B is simply the dilation of A by B, followed by the erosion of the result by B

Opening and Closing

- Opening smoothens the contour of an object, breaks narrow isthmuses, and eliminates thin protrusions
- Closing also tends to smooth sections of contours but, as opposed to opening, it generally fuses narrow breaks and long thin gulfs, eliminates small holes, and fills gaps in the contour

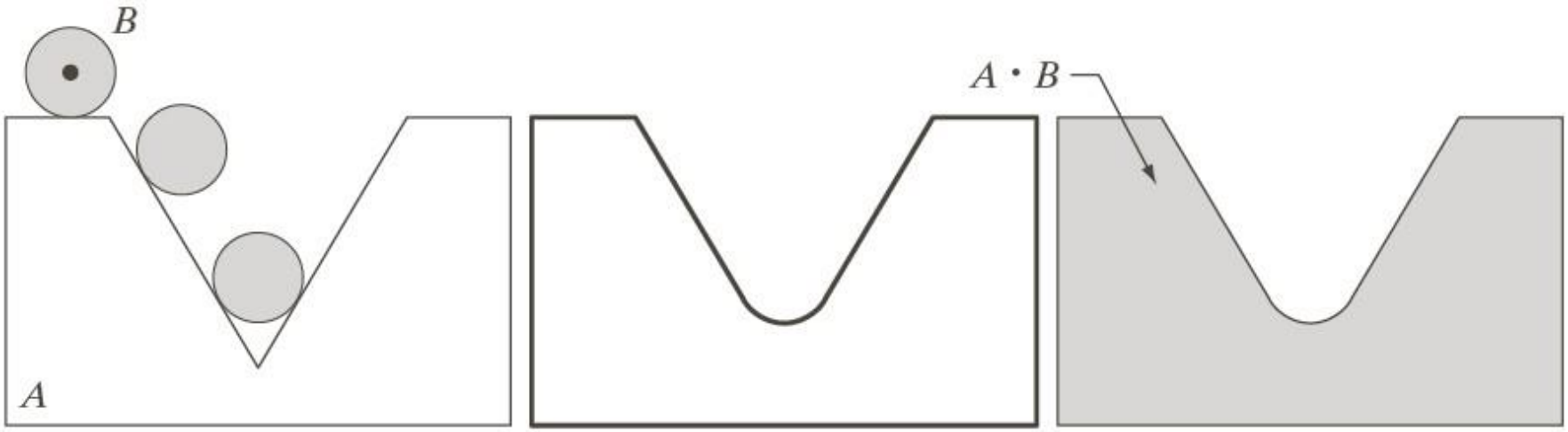
Examples: Opening



a b c d

FIGURE 9.8 (a) Structuring element B "rolling" along the inner boundary of A (the dot indicates the origin of B). (b) Structuring element. (c) The heavy line is the outer boundary of the opening. (d) Complete opening (shaded). We did not shade A in (a) for clarity.

Examples: Closing



a b c

FIGURE 9.9 (a) Structuring element B "rolling" on the outer boundary of set A. (b) The heavy line is the outer boundary of the closing. (c) Complete closing (shaded). We did not shade A in (a) for clarity.

Opening and Closing



a b
d c
e f

FIGURE 9.11
(a) Noisy image.
(b) Structuring element.
(c) Eroded image.
(d) Opening of A.
(e) Dilation of the opening.
(f) Closing of the opening.
(Original image courtesy of the National Institute of Standards and Technology.)